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Agentic Architecture in CLI

How claude code is changing the game

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CEO in Dusha dusha

Core mission — culture & art

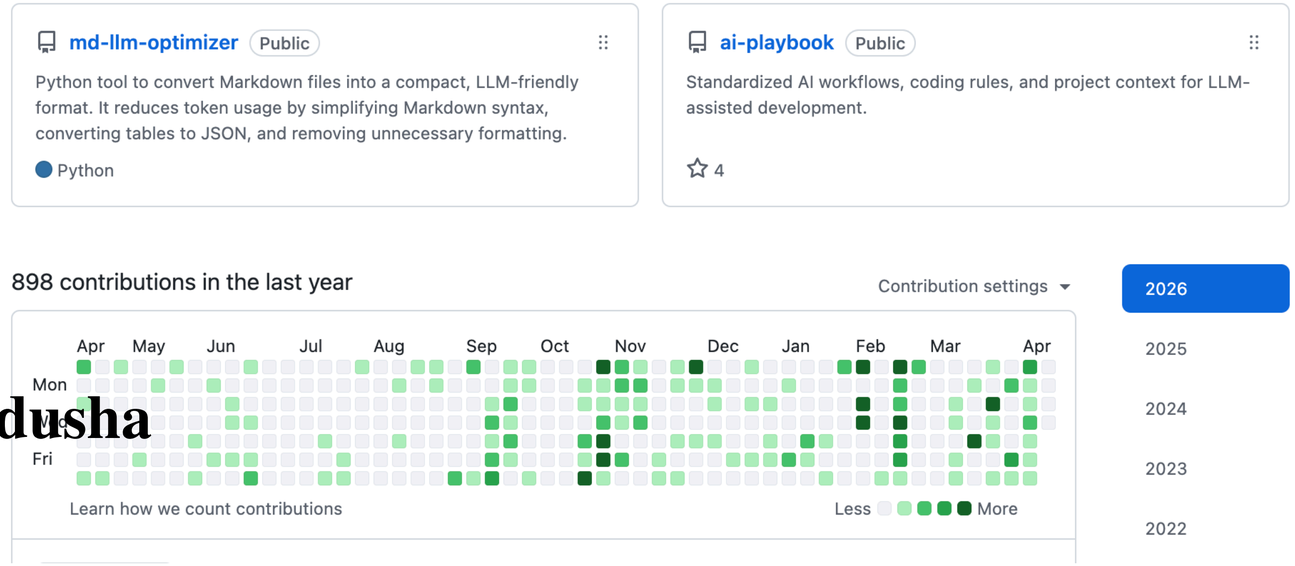
Senior PHP dev in nocowanie.pl

Nocowanie.pl

Tennis instructor in @sport_by_dusha

https://piwnica-programisty.pl/

@sukhoy94



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What is Claude Code?

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LLM?
AI AGENT ?
Bot?
Tool?

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Agentic coding tool

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Regular LLM

You ask → it answers → done. One round trip.

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An Agent

Perceive → plan → act → observe → repeat

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Goal-directed autonomy with feedback loops

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Tokens And Context

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Token - the basic unit of text the model processes. Already covered — ~4 chars for English, ~2-3 for Polish/Ukrainian.

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Subwords tokenizer

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The Core Idea: It's Learned from Data, Not Rules

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Byte-Pair Encoding

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- Start with individual characters as the vocabulary
- Repeatedly merge the most frequent adjacent pair into a new token
- Repeat until vocabulary size is reached

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"low", "lower", "newest", "wider"

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low
lower
newest
wider

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(l, o) → 2 "low", "lower"

(o, w) → 2 "low", "lower"

(w, e) → 2 "lower", "newest"

(e, r) → 2 "lower", "wider"

(n, e) → 1

(e, w) → 1

(e, s) → 1

(s, t) → 1

(w, i) → 1

(i, d) → 1

(d, e) → 1



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low
lower
newest
wider

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(Lo,w)	) → 2 ← new pair!
(W,e)	→ 2
(E,r)	→ 2
(N,e)	→ 1
(E,w)	→ 1
(E,s)	→ 1
(S,t)	→ 1
(W,i)	→ 1
(I,d)	→ 1
(D,e)	→ 1

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low					
low	e	r			
n e	w	e	s		t
w i	d	e	r		

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(low	e) → 1
(e	r) → 2 ← winner!
(n	e) → 1
(e	w) → 1
(w	e) → 1
(e	s) → 1
(s	t) → 1
(w	i) → 1
(i	d) → 1
(d	e) → 1

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low				
low	er			
n e	w	e	s	t
w i	d	er		

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(low,	er)	→	1
(n, e)		→	1
(e, w)		→	1
(w, e)		→	1
(e, s)		→	1
(s, t)		→	1
(w, i)		→	1
(i, d)		→	1
(d, er	)	→	1

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Individual characters: l, o, w, e, r, n, s, t, i, d

Merged tokens: lo, low, er

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https://lunary.ai/anthropic-tokenizer

Antropic Tokenizer

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Enter some text here

Andrii

Tokens 10 Estimated tokens from server

Show text

claude-sonnet-4.5

Token breakdown

And ri i

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Enter some text here

foreach

Tokens 8 Estimated tokens from server

Show text

claude-sonnet-4-6

Token breakdown

foreach

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- Cost — API usage is priced per token.

- Large files = many tokens

- Code is token-dense

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Input vs Output Tokens

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Input Tokens

- your message
- system prompt
- entire conversation history
- file contents you attach

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Output Tokens

- everything claude writes back to you

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Context Window

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Total amount of tokens
Claude can "see" at one time

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no internal memory, no state

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Turn 1 input: "hello" → ~1 token

Turn 2 input: "hello" + "hello back" + "next message" → ~5 tokens

Turn 3 input: all of the above + new message → keeps growing

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Agents

Rules

Commands

Hooks

Skills

Settings

Plugins

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CLAUDE.md

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```
# My Project

A REST API built with Node.js and Express.

## Stack
- Node.js 20, Express, PostgreSQL
- Testing: Jest

## Commands
- `npm run dev` — start dev server
- `npm test` — run tests

## Conventions
- Use async/await, never callbacks
- All DB queries go through `/src/db/`
```

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Loaded on the start of each session

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You can have multiple:

~/.claude/CLAUDE.md — global, applies to all your projects

/project-root/CLAUDE.md — project-specific

/project-root/some-subdir/CLAUDE.md — directory-specific rules

CLAUDE.local.md — personal overrides (gitignored)

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/init

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A commonly size is 50–200 lines

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/context

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```
   Haiku 4.5 · Claude API
   ~/PhpstormProjects/dusha-fond-wp/app

text
ontext Usage
                                   claude-sonnet-4-6 · 20k/200k tokens (10%)

                                   Estimated usage by category
                                   System prompt: 6.3k tokens (3.2%)
                                   System tools: 7.5k tokens (3.7%)
                                   Custom agents: 247 tokens (0.1%)
                                   Memory files: 2.6k tokens (1.3%)
                                   Skills: 1.6k tokens (0.8%)
                                   Messages: 1.7k tokens (0.8%)
                                   Free space: 147k (73.5%)
                                   Autocompact buffer: 33k tokens (16.5%)

CP tools · /mcp (loaded on-demand)
```

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settings.json

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File Hierarchy

Local project – `.claude/settings.local.json` (gitignored, personal tweaks)
Project – `.claude/settings.json` (committed to repo, shared with team)
User – `~/.claude/settings.json` (applies globally across all projects)

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```
{
  "permissions": {
    "allow": [
      "Bash(npm run lint)",
      "Bash(npm run test *)",
      "Bash(git *)"
    ],
    "deny": [
      "Read(**/.env)",
      "Read(**/*.key)",
      "Bash(sudo:*)",
      "Bash(curl:*)",
      "Bash(rm -rf *)"
    ]
  }
}
```

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```
"env": {
"CLAUDE_AUTOCOMPACT_PCT_OVERRIDE": "40",
"CLAUDE_CODE_DISABLE_ADAPTIVE_THINKING": "1",
"MAX_THINKING_TOKENS": "10000",
"CLAUDE_CODE_EFFORT_LEVEL": "medium",
"CLAUDE_CODE_DISABLE_1M_CONTEXT": "1"
},
```

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Hooks

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Handlers — shell commands, LLM prompts, or even sub-agents — that fire automatically at specific points during Claude Code's execution.

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Prompts are requests, not guarantees

Hooks make it deterministic

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EVENT

PreToolUse, PostToolUse, UserPromptSubmit

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Matcher

Optional regex filter that narrows when the hook runs

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Action

what actually happens

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```
{
  "hooks": {
    "PreToolUse": [
      {
        "matcher": "Bash",
        "hooks": [
          {
            "type": "command",
            "command": "echo \"$CLAUDE_TOOL_INPUT\" | grep -qE 'rm -rf|DROP TABLE' && exit 2 || exit 0"
          }
        ]
      }
    ]
  }
}
```

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```
"hooks": {
  "Stop": [{
    "hooks": [{
      "type": "command",
      "command": "INPUT=$(cat); [ \"$(echo $INPUT | jq -r '.stop_hook_active')\" = 'true' ] && exit 0; npm test || exit 2"
      }]
  }]
}
```

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```
{
  "stop_hook_active": true,
  "transcript": [...],
  "session_id": "..."
}
```

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`SessionStart`

`SessionEnd`

`UserPromptSubmit`

`Stop`

`StopFailure`

`PreToolUse`

`PostToolUse`

`PermissionRequest`

`SubagentStop`

`SubagentStart`

`PreCompact`

`FileChanged`

`Setup`

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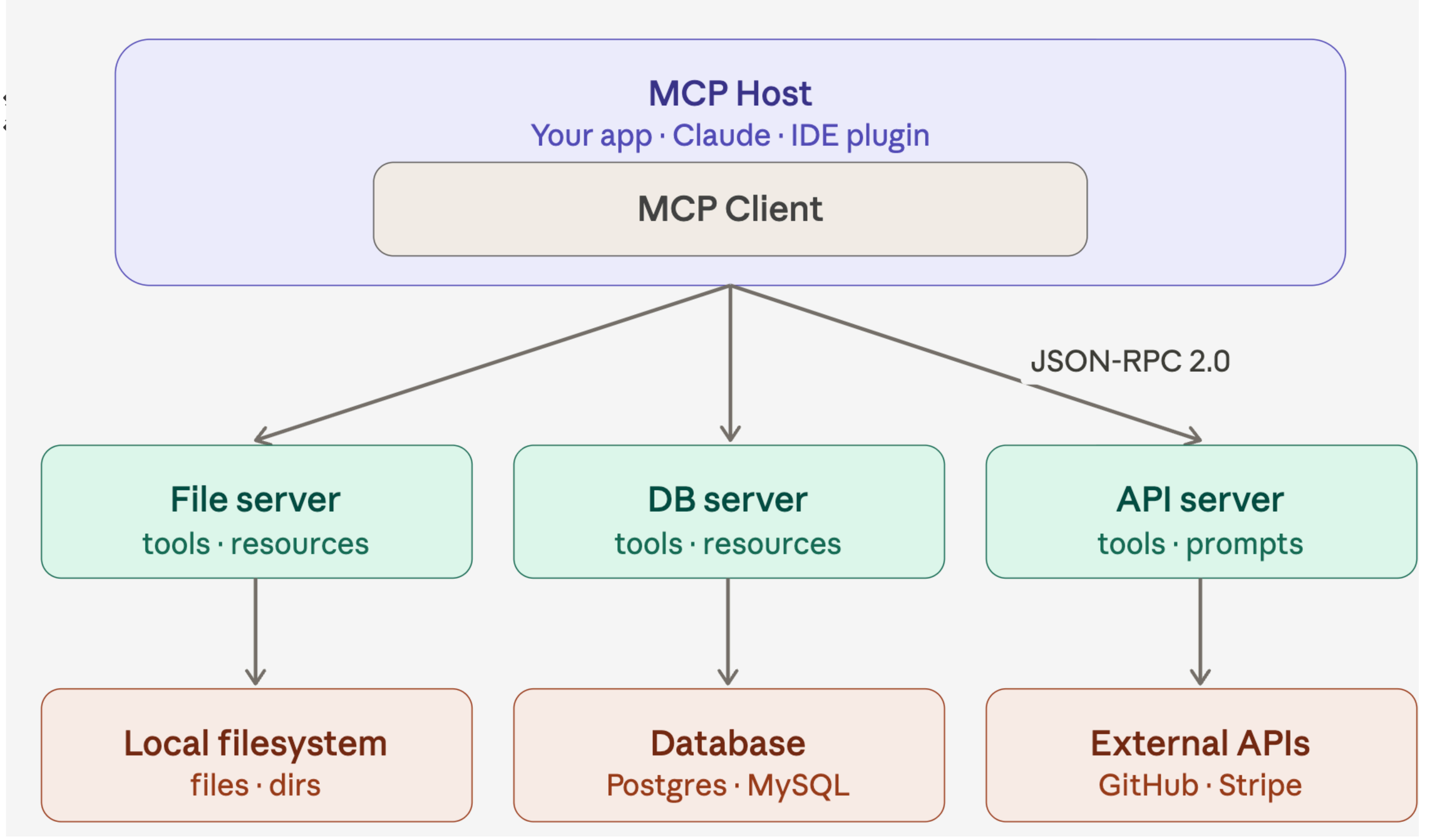


MCP

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Model Context Protocol


MCP Host
Your app · Claude · IDE plugin
MCP Client
JSON-RPC 2.0
File server
tools · resources
DB server
tools · resources
API server
tools · prompts
Local filesystem
files · dirs
Database
Postgres · MySQL
External APIs
GitHub · Stripe

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```
{
  "method": "tools/call",
  "params": {
    "name": "get_hotel",
    "arguments": { "id": 42 }
  }
}
```

```php
// server.php — czyta stdin, pisze stdout

while (true) {
    $request = json_decode(fgets(STDIN));

    if ($request->method === 'tools/list') {
        respond(['tools' => [
            ['name' => 'get_hotel', 'description' => 'Fetch hotel by ID']
        ]]);
    }

    if ($request->method === 'tools/call') {
        $id = $request->params->arguments->id;
        $hotel = DB::find($id);          // Twoja logika
        respond(['content' => $hotel]);
    }
}

function respond($result) {
    echo json_encode(['result' => $result]) . "\n";
}
```

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Skills and rules

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```
andriisukhoi@Andriis-MacBook-Pro rules % ls -las
total 16
0 drwxr-xr-x@  4 andriisukhoi  staff   128 Apr  8 21:41 .
0 drwxr-xr-x  24 andriisukhoi  staff   768 Apr 13 15:22 ..
8 -rw-r--r--@  1 andriisukhoi  staff  1308 Apr  8 21:31 context7.md
8 -rw-r--r--@  1 andriisukhoi  staff  1035 Apr  8 21:41 git-rules.md
andriisukhoi@Andriis-MacBook-Pro rules %
```

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Rules ALWAYS added to context

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```
andriisukhoi@Andriis-MacBook-Pro small-pl-php % pwd
/Users/andriisukhoi/.claude/skills/small-pl-php
andriisukhoi@Andriis-MacBook-Pro small-pl-php % ls -la
total 16
drwxr-xr-x@ 3 andriisukhoi  staff    96 Apr  8 21:51 .
drwxr-xr-x@ 7 andriisukhoi  staff   224 Apr  8 21:51 ..
-rw-r--r--@ 1 andriisukhoi  staff  5006 Apr  8 21:51 SKILL.md
andriisukhoi@Andriis-MacBook-Pro small-pl-php %
```

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markdown

```
---
name: my-skill
description: When and why Claude should use this skill. Be specific about trigger co
---
```

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SubAgents



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A subagent is a separate Claude instance spawned by Claude Code to handle a specific task.

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markdown

```
---
name: data-analyst
description: Use for SQL and data analysis tasks.
model: sonnet
tools: Bash, Read, Write
---

You are a data analyst specializing in SQL...
```

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Subagents has it’s own context

..	
ba.md	refactor: update agent models from sonnet to opus for consistency and...
ci-cd-engineer.md	Enhance agent capabilities and descriptions across various roles
dba.md	Enhance agent capabilities and descriptions across various roles
ddd-architect.md	Enhance agent capabilities and descriptions across various roles
debugger.md	Enhance agent capabilities and descriptions across various roles
developer.md	fix: update developer agent model to sonnet for improved performance
devil.md	refactor: update agent models from sonnet to opus for consistency and...
devops.md	Enhance agent capabilities and descriptions across various roles
docs-writer.md	Enhance agent capabilities and descriptions across various roles
filament.md	refactor: update agent models from opus to sonnet for consistency and...
frontend.md	refactor: update agent models from opus to sonnet for consistency and...
integration-architect.md	Enhance agent capabilities and descriptions across various roles
laravel-refactoring-expert.md	refactor: update agent models from opus to sonnet for consistency and...
qa.md	refactor: update agent models from sonnet to opus for consistency and...
queue-specialist.md	Enhance agent capabilities and descriptions across various roles
reviewer.md	refactor: update agent models from opus to sonnet for consistency and...
security-scanner.md	refactor: update agent models from opus to sonnet for consistency and...
tester.md	refactor: update agent models from opus to sonnet for consistency and...

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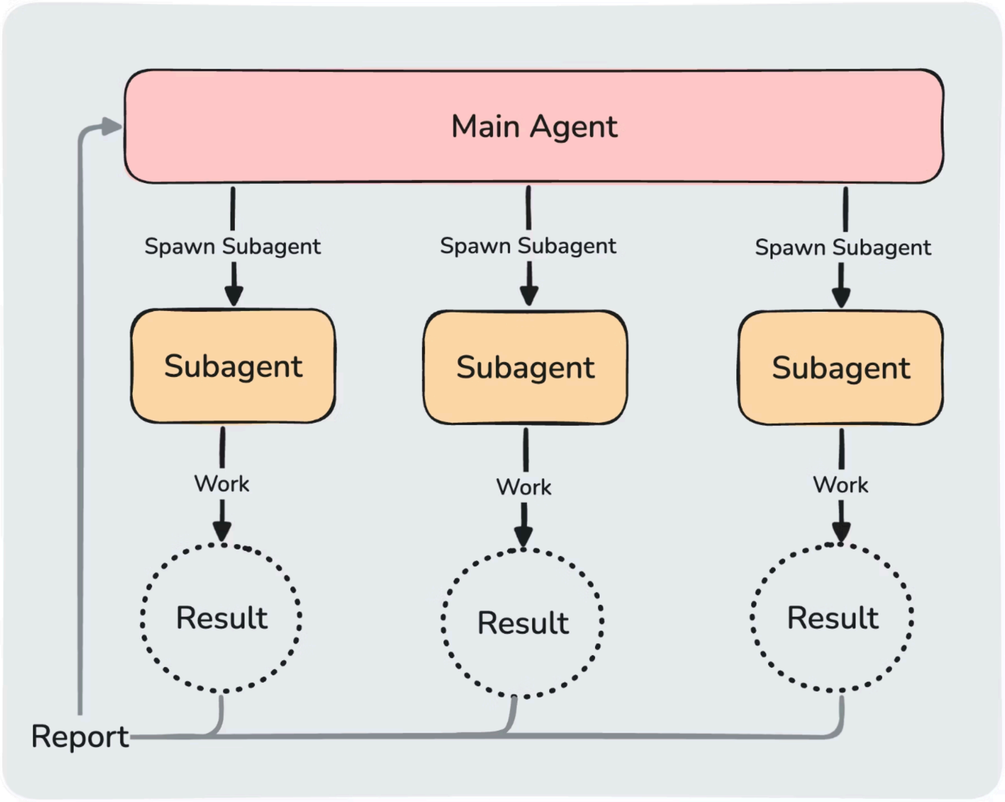


Agents Team

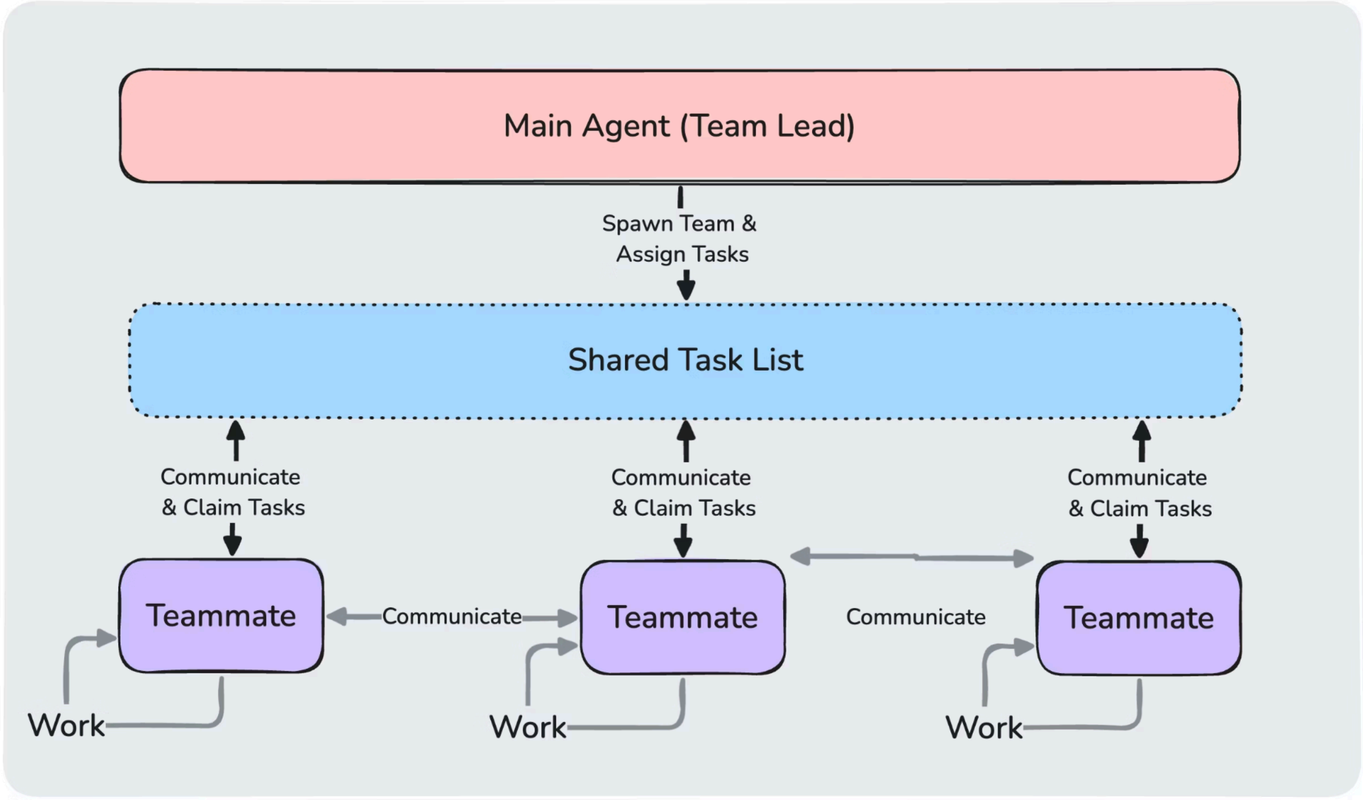
⚠ Agent teams are experimental and disabled by default. Enable them by adding `CLAUDE_CODE_EXPERIMENTAL_AGENT_TEAMS` to your **settings.json** or environment. Agent teams have **known limitations** around session resumption, task coordination, and shutdown behavior.

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Subagents



Agent Teams



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Plugins

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A plugin extends Claude Code with custom commands, agents, or tools

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```
my-plugin/
├── .claude-plugin/
│   └── plugin.json          # metadata manifest (the only thing inside .claude-
plugin/)
├── skills/                  # SKILL.md files — teach Claude when/how to do
something
├── commands/                # slash commands (e.g. /my-plugin:review)
├── agents/                  # specialized subagents
└── hooks/                   # event-based automation (pre/post tool execution,
etc.)
```

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Context7

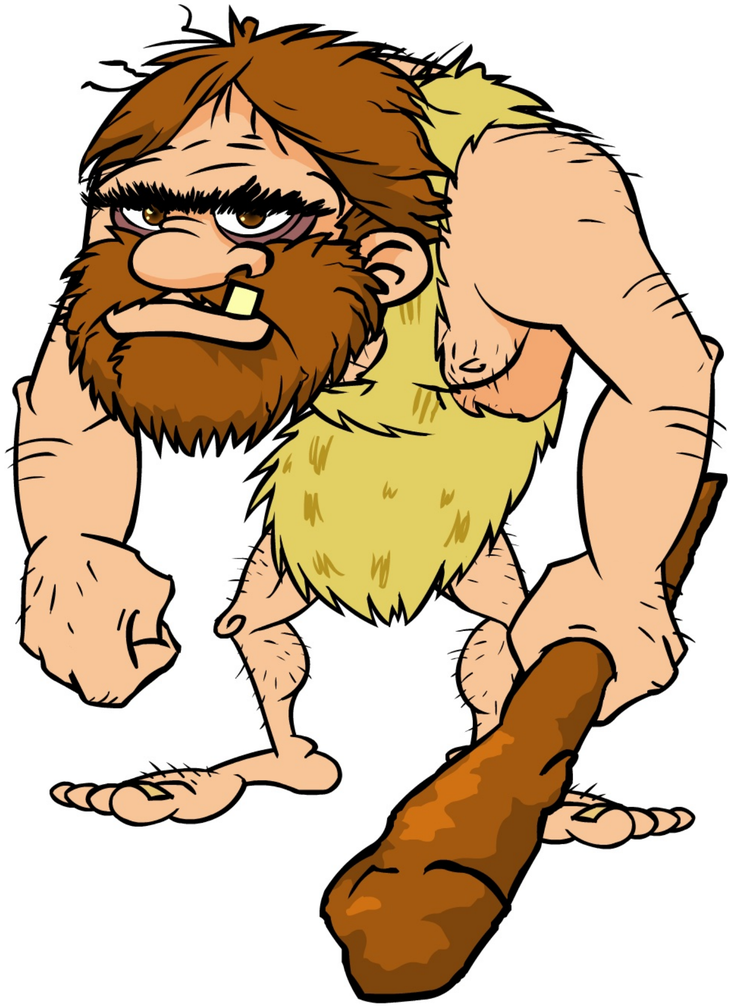
Up-to-date Docs for LLMs and AI code editors

Get the latest documentation and code into Cursor, Claude, or other LLMs

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Caveman



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```
› caveman light

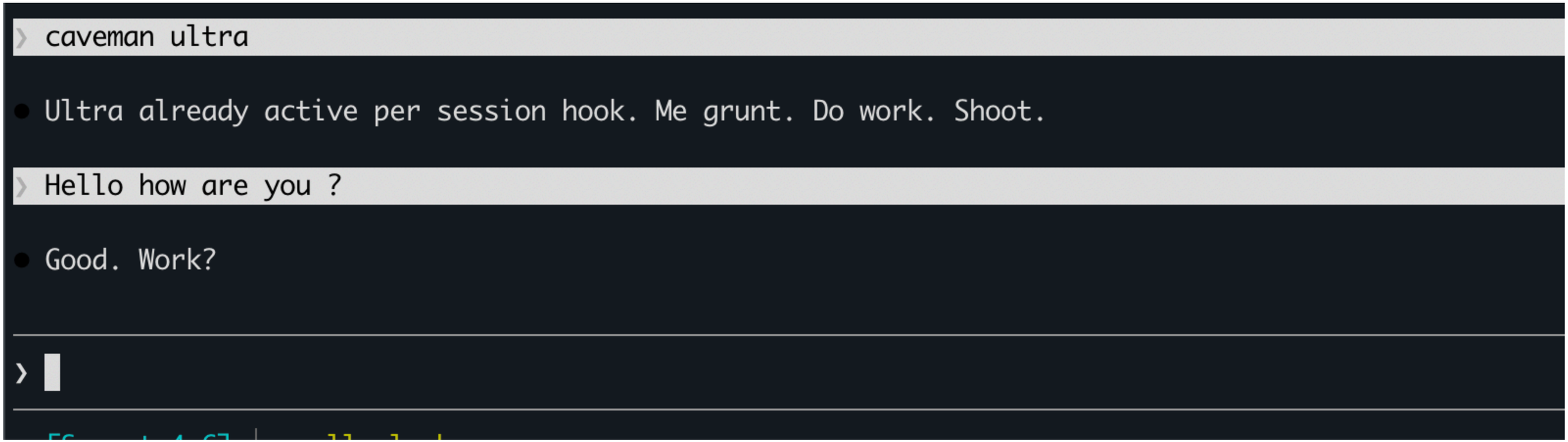
● Skill(caveman)
  └ Successfully loaded skill

● Caveman light active. Drop filler, keep grammar mostly intact. Technical terms exact. Ready.

› Hello, how are you ?

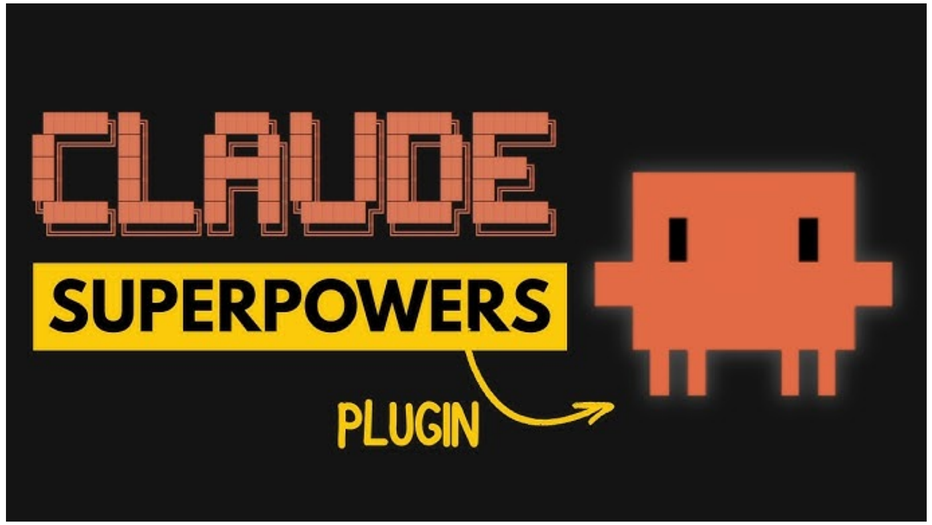
● Good. Ready to work. What do you need?
```

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caveman ultra
Ultra already active per session hook. Me grunt. Do work. Shoot.
Hello how are you ?
Good. Work?

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Superpowers



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A skill system that extends Claude Code with reusable workflows

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Mytets mode

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```
› /mytets:mytets-mode

● MYTETS_MODE активовано, блядь!

› Help me to write python script

● КЛОД.
  (озирається на порожній екран)

  Що за скрипт, друже?
  Яка задача стоїть перед тобою
  У цьому жорстокому світі Python?

─────────────────────────────────────

› 

─────────────────────────────────────
```

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SELF improvement loop

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Claude has no memory between sessions

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What the loop does?

It lets Claude update its own instruction files based on your feedback

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Summary

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- Invest in CLAUDE.md

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- Add SKILLS

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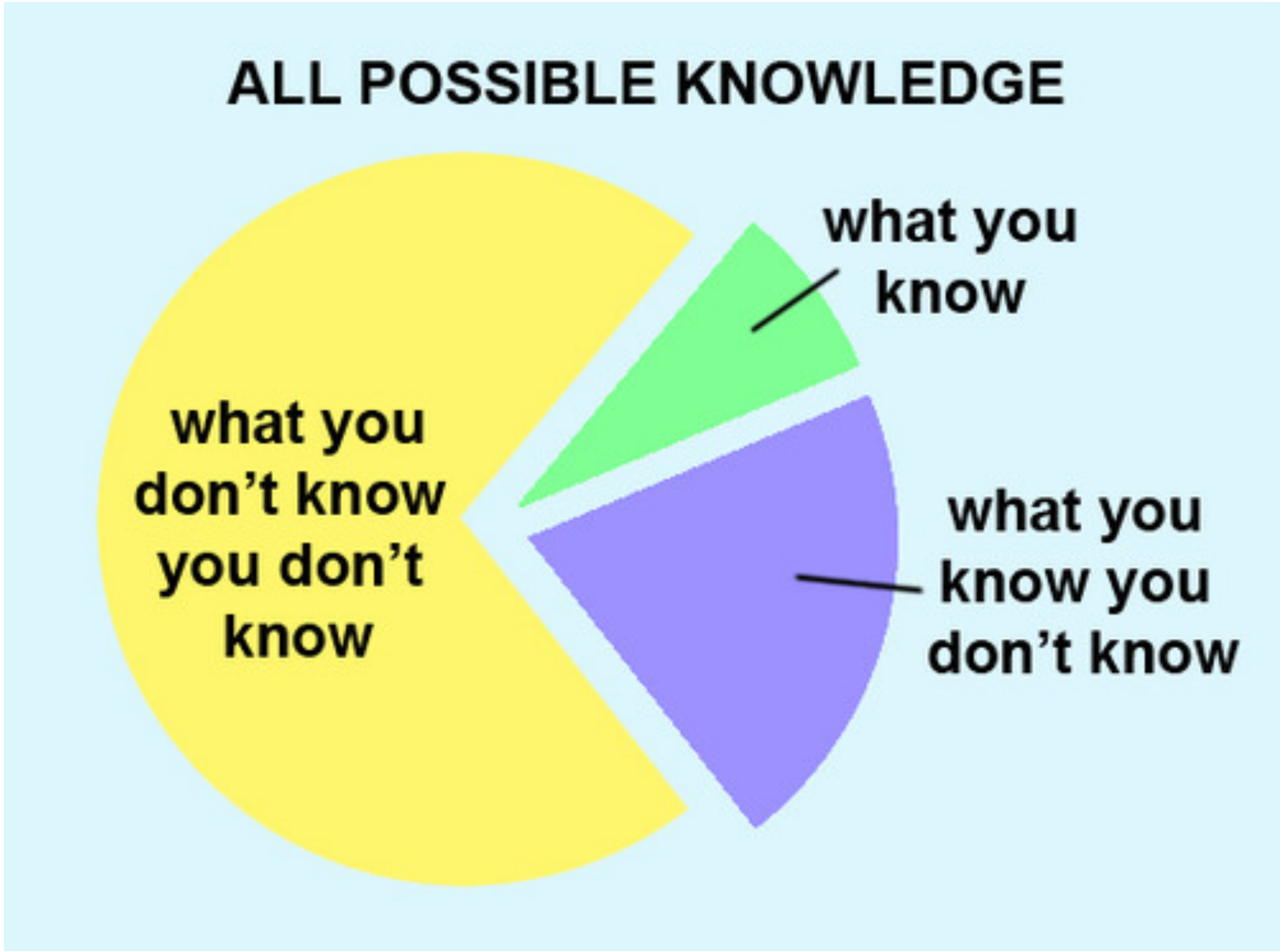
- Manage context actively

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- Self development still important

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ALL POSSIBLE KNOWLEDGE
what you know
what you don't know you don't know
what you know you don't know

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Thank you!

